

Trade Tariffs and Exchange Rates in a Multilateral World

Theoretical derivations

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1 Introduction

In a two-country model, an import tariff appreciates the tariffing country's currency under the Marshall–Lerner condition. In a multilateral setting, a discriminatory tariff also shifts expenditure across foreign suppliers. As a result, the tariffing country's bilateral exchange rates need not all move in the same direction.

These notes characterize the exchange-rate effects of tariffs in an N -country model. The general log-linearized equilibrium response is $(-J)^{-1}F$, where F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. Under symmetry and a multilateral Marshall–Lerner condition on J , the tariffing country always appreciates against the target, but its bilateral response against an untariffed bystander is ambiguous and depends on whether the cross-origin elasticity of substitution across foreign imported tradables exceeds the threshold $N\rho_1^*$.

Despite this bilateral ambiguity, the tariffing country's nominal effective exchange rate (NEER) appreciates for every admissible cross-origin elasticity and every N . Under symmetry the two results are the two components of a single decomposition: any tariff vector imposed by the tariffing country splits into a scale component, which alone determines the effective rate, and a discriminatory component, which alone determines the distribution of bilateral rates around it (Section 7). Proofs of all formal results are collected in Appendix D.

2 N -Country Model and Equilibrium

2.1 Production and preferences

Countries are indexed by $i, j \in \{1, \dots, N\}$. We use A for the tariff imposer, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i, \quad (1)$$

with perfect competition and producer-price normalization $P_{T_i} = 1$ in i 's currency, so

$$w_i = A_{T_i}, \quad P_{N_i} = A_{T_i}/A_{N_i}. \quad (2)$$

Wages are constant in own currency, so relative adjustment occurs through exchange rates. Preferences are nested CES with share-linear weights:

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} \quad (\text{tradable share } \alpha_T) \quad (3)$$

$$C_{T_i} = \left[\alpha_D^{1/\eta} C_{T_{ii}}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (\text{home weight } \alpha_D, \text{ elasticity } \eta) \quad (4)$$

$$M_i = \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_{ji}}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{origin weights } \beta_{ij}, \text{ elasticity } \rho). \quad (5)$$

The elasticity η governs substitution between domestic and imported tradables, while ρ governs substitution across foreign origins conditional on importing. The latter margin is inactive at $N = 2$ and active at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias α_D preserves an asymmetry between home and foreign tradables. Full symmetry between home and foreign tradables would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

2.2 Tariffs, prices, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem and are levied by i on imports from j , with $\tau_{ii} = 0$. Consumer prices in i 's currency are $p_{ij} = (1 + \tau_{ij}) e_{ij}$ and $p_{ii} = 1$. The corresponding price indices are

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1-\alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (6)$$

Expenditure shares of income are

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad s_{ij} = \alpha_T (1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i, \quad (7)$$

so $s_{ij} I_i$ is tariff-inclusive spending at consumer prices and $s_{ij} I_i / (1 + \tau_{ij})$ is the tariff-exclusive border value. Tariff revenue is rebated lump-sum:

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (8)$$

Equation (8) determines income in i 's own currency. To add bilateral flows across countries, we express them in a common currency. Taking A 's currency as the numeraire, define common-currency income as $\tilde{I}_i \equiv e_{Ai} I_i$. Expenditure shares are ratios of same-currency values and are therefore unchanged by this conversion. Since the tariff wedge is a domestic transfer, trade balances are valued at producer prices in the common currency:

$$\text{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \quad f_{ij} \equiv \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \quad (9)$$

where f_{ij} is the border value, in A 's currency, of i 's imports from j .

By Walras' law, only $N - 1$ trade-balance conditions are independent. We also assume no internationally traded assets, so $\text{TB}_i = 0$ for all i . Given $\{\tau_{ij}\}$, the $N - 1$ free exchange rates therefore solve $\text{TB}_i = 0$ for $i \neq A$.

2.3 Exchange-rate concepts

Define e_{ij} as units of i 's currency per unit of j 's, with $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. A rise in e_{ij} therefore represents a depreciation of i against j . Log changes in bilateral exchange rates relative to A are denoted by $\nu_j \equiv d \log e_{Aj}$, with $\nu_A = 0$. Relative wages, terms of trade, real exchange rates, and effective exchange rates are defined as

$$\omega_{ij} \equiv \frac{e_{ij} w_j}{w_i}, \quad \text{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij} P_{T_j}} = \frac{1}{e_{ij}}, \quad q_{ij} \equiv \frac{e_{ij} P_j}{P_i}, \quad (10)$$

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij} d \log e_{ij}, \quad dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})}, \quad (11)$$

with $w_{ij} = 1/(N - 1)$ in the symmetric model. Under the production normalization,

$$d \log e_{ij} = d \log \omega_{ij} = -d \log \text{ToT}_{ij}, \quad (12)$$

so the nominal exchange-rate results can equivalently be expressed in terms of relative wages or terms of trade.

3 Two Countries

At the two-country benchmark, the equilibrium is a single trade-balance condition, $\text{TB}_A = 0$, in the single free exchange rate e_{AB} . Suppose A imposes $d\tau = dt_{AB}$. Linearizing at the balanced free-trade baseline with normalized incomes, the two flows are $f_{AB} = f_{BA} = m$, with $m \equiv \alpha_T(1 - \alpha_D)$, and we write $\nu_B \equiv d \log e_{AB}$. Each country has a single import origin, so the bilateral consumer price coincides with the import index: $dp_{AB} = d \log P_{M_A} = \nu_B + d\tau$ and $dp_{BA} = d \log P_{M_B} = -\nu_B$. From (6) and (7), $d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}$, so each import share responds only through substitution between home and imported tradables, while incomes respond to the tariff rebate in (8) and, for B , to the valuation of its income in A 's currency:

$$\begin{aligned} d \log s_{AB} &= (1 - \eta) \alpha_D (\nu_B + d\tau), & d \log \tilde{I}_A &= m d\tau, \\ d \log s_{BA} &= -(1 - \eta) \alpha_D \nu_B, & d \log \tilde{I}_B &= \nu_B. \end{aligned}$$

Substituting into (9) gives

$$\begin{aligned} d\text{TB}_A &= m[d\log s_{BA} + \nu_B] - m[d\log s_{AB} + m d\tau - d\tau] \\ &= m[(1 + 2\alpha_D(\eta - 1))\nu_B + (1 - m + \alpha_D(\eta - 1))d\tau]. \end{aligned}$$

The coefficient on ν_B is the trade-balance response to exchange-rate adjustment, and the coefficient on $d\tau$ is the impact disturbance created by the tariff at the unchanged exchange rate. Write $D_2 \equiv 2\alpha_D(\eta - 1) + 1$ for the former and define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \underbrace{\alpha_D \eta}_{\text{home switching}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{expenditure absorption}} > 0. \quad (13)$$

While ρ governs substitution across foreign origins, ρ_1^* captures the opposing effects of substitution toward A 's own tradable and expenditure absorption outside the imported-tradables margin. The cross-origin elasticity ρ is inactive at $N = 2$, since each country has a single foreign supplier; it returns at $N = 3$ as the foil to ρ_1^* , and the thresholds of Sections 5 and 6 compare the strength of foreign-supplier diversion with these opposing forces. In this notation,

$$d\text{TB}_A = \underbrace{m D_2 \nu_B}_{\text{adjustment}} + \underbrace{m \rho_1^* d\tau}_{\text{impact disturbance}} = 0, \quad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \quad (14)$$

and therefore

$$\boxed{\left. \frac{d\log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_1^*}{D_2}} \quad (15)$$

In reduced form, let export volume be $X(1/e)$ and import volume be $M((1 + \tau)e)$, with elasticities ε_X and ε_M . Balanced trade gives $\partial\text{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Comparing this expression with (14) per unit of import value gives

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \quad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1. \quad (16)$$

Proposition 3.1 (Conventional wisdom, $N = 2$). *At $N = 2$, exchange-rate adjustment stabilizes the trade balance if and only if $D_2 > 0$, the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. In our nested-CES model, it holds whenever $\eta \geq 1$. Since the impact disturbance $m\rho_1^*$ is positive for all admissible parameters, a unilateral import tariff appreciates the tariffing country's currency: $d\log e_{AB}/d\tau = -\rho_1^*/D_2 < 0$.*

Appendix A gives the exact finite-tariff Cobb–Douglas solution.

4 From Two to N Countries

The following results require only that trade balances are differentiable with respect to exchange rates and tariffs. They do not depend on the nested-CES structure of our model.

4.1 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N - 1$ independent trade-balance conditions, $TB_i = 0$ for $i \neq A$, as functions of exchange rates and tariffs. Stack the exchange-rate changes as $d\nu \equiv (\nu_j)_{j \neq A}$ and the bilateral tariff changes as the vector $d\tau \equiv (dt_{jk})_{j \neq k}$. The conditions for $i \neq A$ are

$$J d\nu + G d\tau = 0, \quad J_{il} \equiv \frac{\partial dTB_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial dTB_i}{\partial dt_{jk}} \Big|_{\nu \text{ fixed}}. \quad (17)$$

For a particular tariff experiment, write the tariff vector as $d\tau = a d\tau$, where a is a fixed direction vector and $d\tau$ is the scalar size of the tariff change. Defining the impact vector $F \equiv Ga$, the equilibrium conditions become

$$J d\nu = -F d\tau, \quad \frac{d\nu}{d\tau} = (-J)^{-1} F. \quad (18)$$

F gives the trade-balance effects of the tariff at unchanged exchange rates, including substitution and income effects, while J describes how trade balances respond to exchange-rate changes. The equilibrium response $(-J)^{-1} F$ combines these initial disturbances through the exchange-rate adjustment mechanism.

4.2 The multilateral Marshall–Lerner condition

Two structural identities restrict $\tilde{J}$, the full $N \times N$ Jacobian before country A is dropped. First, each trade balance is homogeneous of degree one in currency values. By Euler’s theorem, the log-derivatives in row i sum to TB_i , which is zero at a balanced baseline. The rows of $\tilde{J}$ therefore sum to zero. Second, Walras’ law gives $\sum_i TB_i \equiv 0$, so the columns also sum to zero. Thus,

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i, \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l. \quad (19)$$

At $N = 2$, the system (18) is the scalar condition for B . Since $TB_B = -TB_A$, (14) gives $-J = mD_2$ and $F = -m\rho_1^*$, so exchange-rate adjustment stabilizes the trade balance if and only if $-J > 0$, that is, $D_2 > 0$, and the response $(-J)^{-1} F = -\rho_1^*/D_2$ is then a positive multiple of the impact disturbance. For N countries, the scalar requirement $-J > 0$ becomes the requirement that $-J$ be a nonsingular M-matrix, so that $(-J)^{-1}$ is again nonnegative.

Definition 4.1 (Multilateral Marshall–Lerner condition). *The adjustment mechanism satisfies the multilateral Marshall–Lerner condition if*

$$-J \text{ is a nonsingular M-matrix: } J_{il} \geq 0 \quad (l \neq i) \quad \text{and} \quad (-J)^{-1} \text{ exists with } (-J)^{-1} \geq 0. \quad (20)$$

By (18), each equilibrium exchange-rate response is a nonnegative-weighted combination of the initial trade-balance disturbances. If the components of F have the same sign, a nonnegative $(-J)^{-1}$

cannot produce responses of the opposite sign. Opposite-signed equilibrium responses therefore require a mixed-sign F . When F has mixed signs, the sign of an individual response depends on a weighted combination of the entire disturbance vector, not simply on the corresponding F_i . The condition also implies $\det(-J) > 0$, which we use in the threshold calculations below.¹ At $N = 2$, the condition is $D_2 > 0$, and Section 3 established its exact correspondence with the textbook elasticity condition in our model.

A sufficient condition for the multilateral Marshall–Lerner condition is gross substitutability in the full Jacobian.

Assumption 1 (Gross substitutability). $\tilde{J}_{il} \geq 0$ for all $l \neq i$.

In the coordinates $\nu_l = d \log e_{Al}$, a rise in ν_l is an appreciation of currency l against the numeraire. Gross substitutability therefore requires that one country’s appreciation does not worsen another country’s trade balance. Together with (19), this implies adverse own effects and weak diagonal dominance, which deliver the M-matrix property under the conditions below.

Proposition 4.2 (Sufficiency of gross substitutability). *Under Assumption 1, with J irreducible and $\tilde{J}_{iA} > 0$ for some i , the multilateral Marshall–Lerner condition (20) holds.*

The diagonal-dominance calculation and the remaining M-matrix arguments are in Appendix D. Gross substitutability restricts the local effects in J , while the multilateral Marshall–Lerner condition restricts the equilibrium adjustment mapping $(-J)^{-1}$. The results below require only the latter. Its implication for signs can be stated directly.

Corollary 4.3 (Sign rule). *Under (20),*

$$\text{sign}\left(\frac{d\nu_i}{d\tau}\right) = \text{sign}\left(\sum_j \vartheta_{ij} F_j\right), \quad \vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0. \quad (21)$$

5 The Symmetric Benchmark

This section applies the general framework of Section 4 to the nested-CES model. We first derive the linearized system for general N and the full Jacobian at the symmetric baseline, then the general symmetric N -country responses, with $N = 3$ used to illustrate the distinction between impact and equilibrium responses. Section 4 is model-free, but Sections 5.2–5.3 and Sections 6–8 characterize tariff responses at the *symmetric benchmark* of Section 5.1, with equal bilateral weights and balanced trade. Appendix B relaxes symmetry and examines which results survive.

5.1 The linearized system

Linearizing at a balanced free-trade baseline with normalized incomes, define the home share $h_i = \alpha_D$, within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = m b_{ij}$ with $m \equiv \alpha_T(1 - \alpha_D)$, and flows

¹Given the Z-matrix sign pattern, the condition is further equivalent to $\text{Re } \lambda(J) < 0$, i.e., to local stability of the tatonnement $\dot{\nu}_i = k_i \text{TB}_i$ for any $k_i > 0$, so simple dynamic exchange-rate adjustment processes converge to the equilibrium.

$f_{ij} = s_{ij}\tilde{I}_i$, where $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Thus, $d \log \tilde{I}_i$ includes the valuation term ν_i in addition to any tariff revenue, while wages remain constant in local currency. Perturbing this baseline with tariffs dt_{ij} gives

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \quad (dp_{ii} = 0), \quad (22)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (23)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (24)$$

$$d \log s_{ij} = \underbrace{(1 - \rho)(dp_{ij} - d \log P_{M_i})}_{\text{across origins}} + \underbrace{(1 - \eta)\alpha_D d \log P_{M_i}}_{\text{home vs. imports}}, \quad (25)$$

$$d \log \tilde{I}_i = \nu_i + \sum_{j \neq i} s_{ij} dt_{ij}, \quad (26)$$

$$d \text{TB}_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log \tilde{I}_k - dt_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log \tilde{I}_i - dt_{ij}). \quad (27)$$

The coefficients in (27) give the model-specific entries of J and G in (17).

At the symmetric baseline, Lemma D.1 in Appendix D reduces the general conditions of Section 4 to a single parameter restriction, showing that every off-diagonal entry of the full $N \times N$ Jacobian is the same positive multiple of

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad D_N \equiv N\alpha_D(\eta - 1) + (N-2)\rho + 1. \quad (28)$$

Thus, weak gross substitutability holds if and only if $D_N \geq 0$. The strict inequality $D_N > 0$ implies the multilateral Marshall–Lerner condition (20): every off-diagonal entry is then strictly positive, so the irreducibility and $\tilde{J}_{iA} > 0$ conditions in Proposition 4.2 hold automatically. At $D_N = 0$, weak gross substitutability still holds, but $\tilde{J}$ and J vanish, so we exclude this boundary case. If $D_N < 0$, the off-diagonal entries of J are negative and the sign condition in (20) fails. Hence, under symmetry, $D_N > 0$ is equivalent to the multilateral Marshall–Lerner condition. Since $\rho > 0$, $\eta \geq 1$ is sufficient for $D_N \geq 1 > 0$ for every N .

Symmetry therefore reduces the adjustment mechanism to a fixed matrix scaled by D_N . We refer to D_N as the model's *multilateral trade-balance adjustment coefficient*: its sign determines whether exchange-rate adjustment is stabilizing, while its magnitude governs the strength of that adjustment. In particular,

$$\eta \geq 1 \implies D_N > 0 \iff \text{multilateral Marshall–Lerner condition (20) under symmetry}, \quad (29)$$

with $\eta \geq 1$ sufficient but not necessary. At $N = 2$, Section 3 showed that $D_2 = \varepsilon_X + \varepsilon_M - 1$, so $D_2 > 0$ is exactly the textbook Marshall–Lerner condition. The N -country responses below can therefore be read as tariff-induced reallocation forces divided by the strength of trade-balance adjustment. The numerators combine the components of F , while D_N scales the common denominator. Because

$(-J)^{-1}$ combines the entire disturbance vector, an individual F_i does not by itself determine the sign of the corresponding exchange-rate response.

5.2 General symmetric N

Now suppose A imposes $d\tau = dt_{AB}$ at the symmetric N -country benchmark. The bilateral weights are $1/(N-1)$ and the $N-2$ bystanders are identical, with common exchange-rate response ν_C . Equations (22)–(23) give

$$d\log P_{MA} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \quad d\log P_{MB} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \quad d\log P_{MC} = \frac{\nu_B - 2\nu_C}{N-1}, \quad (30)$$

with $d\log \tilde{I}_A = \frac{m}{N-1}d\tau$, $d\log \tilde{I}_B = \nu_B$, and $d\log \tilde{I}_C = \nu_C$. Collecting (27) for B and a representative bystander, and scaling the rows by $(N-1)^2/m$, gives

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \quad D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1. \quad (31)$$

The counting matrix has determinant N , and ρ_1^* is the two-country impact coefficient (13), which recurs for every N . Inverting the system gives

$$\frac{d\log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{N D_N} < 0, \quad \frac{d\log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N D_N}, \quad \frac{d\log e_{BC}}{d\tau} = \frac{(N-1)\rho}{N D_N} > 0. \quad (32)$$

These expressions reduce to (15) at $N=2$. The next subsection uses $N=3$ to distinguish the initial trade-balance effects from the equilibrium exchange-rate responses.

5.3 Three countries: impact versus equilibrium thresholds

With $N=3$, $\beta_{ij} = \frac{1}{2}$ and $s_{ij} = m/2$. Collecting (27) for B and C gives the $N=3$ case of (31) in disturbance-adjustment form:

$$J = -\frac{mD_3}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \quad D_3 = 3\alpha_D(\eta-1) + \rho + 1. \quad (33)$$

The initial disturbances are $F_B = -\frac{m}{4}(\rho + \rho_1^*) < 0$ and $F_C = \frac{m}{4}(\rho - \rho_1^*)$. The target B therefore always moves toward deficit. For the bystander C , trade diversion raises exports, while lower income in B and substitution toward A 's home tradable work in the opposite direction. The net effect on C is positive if and only if $\rho > \rho_1^*$. Inverting the system gives

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1} F d\tau = -\frac{1}{3D_3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D_3} \begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \quad (34)$$

Proposition 5.1 (Three-country threshold). *With $D_3 > 0$,*

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}} \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0. \quad (35)$$

A always appreciates against B, B always depreciates against C, and A depreciates against the untariffed C if and only if $\rho > \rho_3^ \equiv 3\rho_1^*$.*

The condition $F_C > 0$, or $\rho > \rho_1^*$, says that C benefits at unchanged exchange rates; it does not imply that the A – C bilateral rate reverses. By the sign rule, $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_1^*$. Thus, C 's equilibrium exchange rate depends not only on its own diversion gain but also on B 's deterioration. Reversal requires the former to be large enough to outweigh the latter.

Corollary 5.2 (Single elasticity). *With $\rho = \eta = \sigma$, $\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T) \geq 0$ whenever $\alpha_D \geq 1/3$, for any α_T , with equality only at $\alpha_D = 1/3$, $\alpha_T = 1$. The reversal is therefore impossible. At the equal-weights point $\alpha_D = 1/3$, $\alpha_T = 3/4$, $d \log e_{AC}/d\tau = -1/(12\sigma) \rightarrow 0^-$.*

When domestic and foreign tradables enter a single CES nest, the diversion gain therefore cannot offset B 's deterioration when $\alpha_D \geq 1/3$. More generally, $3\rho_1^* \geq \eta$ whenever $\alpha_D \geq 1/3$, so a bilateral depreciation of A against C requires $\rho > \eta$. A cross-origin elasticity above the home–import elasticity is therefore necessary, but not sufficient.

6 Bilateral Ambiguity and Effective-Rate Preservation

Two main results follow from (32). First, bilateral exchange rates need not all move in the same direction.

Theorem 6.1 (Dilution threshold). *Under (20), the bystander bilateral rate reverses if and only if*

$$\boxed{\rho > \rho_N^* = N\rho_1^*}. \quad (36)$$

The factor N is an adjustment fact, not an impact fact. At unchanged exchange rates the bystander's disturbance has the sign of $\rho - \rho_1^*$ for every N (the second row of (31), and F_C in (33) at $N = 3$), so adding foreign suppliers does not spread trade diversion thinner at impact. By Lemma D.1, however, the reduced Jacobian is $-J = c(NI - \mathbf{1}\mathbf{1}')$ with $c = mD_N/(N - 1)^2$, so its zero-sum (discriminatory) direction has eigenvalue Nc against c for the uniform direction: zero-sum disturbances are absorbed N times faster by exchange-rate adjustment than uniform ones. The diversion disturbance must therefore be N times the opposing uniform disturbance before an individual bilateral rate reverses. In the nested-CES model, the resulting threshold is exactly $N\rho_1^*$.² With $\rho = \eta = \sigma$, however, $\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0$ whenever $\alpha_D \geq 1/N$.

²The exact linearity in N is a symmetric nested-CES result. At any balanced baseline where A and a bystander both import from both foreign origins, sufficiently strong cross-origin substitution still reverses the bystander rate, provided the multilateral Marshall–Lerner condition continues to hold (Appendix B). Appendix B also shows that the NEER cancellation below survives to leading order within a structured asymmetric family but can reverse for a sufficiently under-weighted target, and Appendix C collects the correspondence across N .

The reversal is therefore impossible in the home-biased region when the two substitution margins share a common elasticity.

The second main result is that the effective rate cannot reverse. Averaging (32) with symmetric weights gives

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \boxed{-\frac{\rho_1^*}{D_N}} < 0, \quad \nu_B - \nu_C = -\frac{(N-1)\rho}{ND_N} d\tau. \quad (37)$$

Theorem 6.2 (NEER preservation). *Under symmetry, a unilateral tariff appreciates the tariffing country's NEER for every $N \geq 2$ and every ρ satisfying (20).*

Cross-origin substitution can reverse the response of an individual bystander rate because it reallocates expenditure across foreign suppliers. Under symmetry, these reallocation effects cancel from the NEER numerator in (37) and therefore cannot reverse the aggregate response. This does not make the magnitude of the NEER response independent of ρ , since ρ still enters D_N .

Substitution between home and foreign goods determines the sign of the effective response, while substitution across foreign suppliers redistributes adjustment across bilateral rates and can reverse some of them. Section 7 makes the separation exact: the two margins are the two eigen-directions of the symmetric response operator, so that any tariff vector splits into a scale component that alone moves the effective rate and a discriminatory component that alone moves bilateral rates around it.

7 Scale and Discrimination in General Tariff Vectors

Sections 5 and 6 analyze a single bilateral tariff. This section characterizes the response to an arbitrary tariff vector imposed by A and shows that, under symmetry, Theorems 6.1 and 6.2 are two components of one decomposition.

7.1 The decomposition

Let $t = (t_j)_{j \neq A}$ collect A 's tariff changes, $t_j \equiv dt_{Aj}$, and split it into a scale component and a discriminatory component,

$$t_j = \bar{t} + \tilde{t}_j, \quad \bar{t} \equiv \frac{1}{N-1} \sum_{j \neq A} t_j, \quad \sum_{j \neq A} \tilde{t}_j = 0. \quad (38)$$

The scale component is the average tariff A levies on its partners, and the discriminatory component records how far each partner's treatment departs from that average. By linearity and the symmetry of the partners, the response to t is the sum of the unilateral responses in (32): each bilateral rate responds to its own tariff with the own-target coefficient and to every other partner's tariff with the bystander coefficient,

$$\nu_j = \alpha t_j + \beta \sum_{k \neq j, A} t_k, \quad \alpha \equiv -\frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad \beta \equiv \frac{\rho - N\rho_1^*}{ND_N}. \quad (39)$$

In matrix form $\nu = R t$ with $R = (\alpha - \beta)I + \beta \mathbf{1}\mathbf{1}'$. The constant vector and its zero-sum complement are the eigenvectors of R , with eigenvalues

$$\lambda_S \equiv \alpha + (N - 2)\beta = -\frac{(N - 1)\rho_1^*}{D_N}, \quad \lambda_D \equiv \alpha - \beta = -\frac{(N - 1)\rho}{ND_N}. \quad (40)$$

Theorem 7.1 (Scale–discrimination decomposition). *At the symmetric benchmark, under (20), the response of A 's bilateral rate against any partner j to any tariff vector t imposed by A is the sum of a scale term, common to all partners and proportional to the average tariff, and a relative-treatment term, proportional to the gap between j 's tariff and that average:*

$$\nu_j = \lambda_S \bar{t} + \lambda_D \tilde{t}_j, \quad d\nu_A^E = \lambda_S \bar{t}, \quad \nu_j - d\nu_A^E = \lambda_D \tilde{t}_j, \quad d \log e_{jk} = \lambda_D (t_k - t_j) \quad (j, k \neq A). \quad (41)$$

The effective rate depends only on the scale component, bilateral deviations from the effective rate depend only on the discriminatory component, and cross rates between partners depend only on tariff differences.

The parameters separate with the components. ρ_1^* enters only λ_S : home switching and expenditure absorption respond to the overall level of protection and move every bilateral rate equally. ρ enters only λ_D : cross-origin substitution responds only to differences in treatment across partners and, being a pure reallocation, averages to zero. D_N scales both, so the adjustment coefficient governs the magnitude of every response without affecting the comparison between the two components. Under (20) both eigenvalues are negative. The scale component appreciates A against every partner, while the discriminatory component appreciates A against partners taxed above the average ($\tilde{t}_j > 0$) and depreciates it against partners taxed below the average ($\tilde{t}_j < 0$).

The sign of any individual bilateral response therefore turns on two numbers, the average tariff and the partner's deviation from it.

Corollary 7.2 (Relative treatment). *Under the conditions of Theorem 7.1, with $\bar{t} > 0$: (i) bilateral responses are ordered by the tariffs, $t_j > t_k$ implies $\nu_j < \nu_k$, so A appreciates most against the partners it taxes most; (ii) A depreciates against partner j if and only if*

$$\frac{\rho}{N} (\bar{t} - t_j) > \rho_1^* \bar{t} \quad \Longleftrightarrow \quad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}, \quad (42)$$

which requires $\rho > N\rho_1^$ and a tariff on j below the fraction $1 - N\rho_1^*/\rho$ of the average.*

Consider A tariffing every partner, but by varying amounts. Every bilateral rate then carries the same scale appreciation $\lambda_S \bar{t}$, and the ranking of responses across partners is fixed by the tariffs alone. A partner taxed above the average sees A appreciate by more than the effective rate. A partner taxed below the average sees a smaller appreciation, and A depreciates against it only if its tariff is low enough relative to the average to satisfy (42). The untaxed bystander of Theorem 6.1 is the case $t_j = 0$, for which (42) reduces to $\rho > N\rho_1^*$. For a particular partner C , the response is $\nu_C = \lambda_S \bar{t} + \lambda_D (t_C - \bar{t})$: the A – C bilateral rate is pinned down by the scale of the policy and by where C sits in the distribution of tariffs, and by nothing else. Quantifying a bilateral response

under a general tariff vector thus requires the average tariff, the partner's deviation from it, and the two eigenvalues.

The decomposition also covers tariffs on several partners at once.

Corollary 7.3 (Number of targets). *If A raises its tariff by τ on a set of k partners, $1 \leq k \leq N-1$, then for every untargeted partner C and every targeted partner B ,*

$$\frac{d \log e_{AC}}{\tau} = \frac{k(\rho - N\rho_1^*)}{ND_N}, \quad \frac{d \log e_{AB}}{\tau} = -\frac{k\rho_1^*}{D_N} - \frac{(N-1-k)\rho}{ND_N}, \quad \frac{d\nu_A^E}{\tau} = -\frac{k\rho_1^*}{D_N}. \quad (43)$$

The bystander reversal threshold is $N\rho_1^*$ for every $k \leq N-2$.

Targeting more partners scales both components of the bystander response by the same factor k , so the threshold is unchanged and only the magnitude grows. At $k = N-1$ the discriminatory component vanishes and the tariff is pure scale,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j. \quad (44)$$

Because all foreign origins face the same tariff, there is no cross-origin reallocation. The elasticity ρ drops out of the numerator, every bilateral rate appreciates by the same amount, and the bilateral and effective responses coincide. The bilateral ambiguity under a discriminatory tariff therefore comes from reallocation across foreign suppliers rather than from protection itself.

7.2 The unilateral results as components

A unilateral tariff $t_B = d\tau$ has $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1} d\tau$, and $\tilde{t}_C = -d\tau/(N-1)$ for every bystander. Substituting into (41),

$$\frac{d \log e_{AC}}{d\tau} = \underbrace{-\frac{\rho_1^*}{D_N}}_{\text{scale}} + \underbrace{\frac{\rho}{ND_N}}_{\text{discrimination}} = \frac{\rho - N\rho_1^*}{ND_N}, \quad \frac{d \log e_{AB}}{d\tau} = -\frac{\rho_1^*}{D_N} - \frac{(N-2)\rho}{ND_N}, \quad \frac{d\nu_A^E}{d\tau} = -\frac{\rho_1^*}{D_N}. \quad (45)$$

Theorem 6.2 is the projection onto the scale direction, $d\nu_A^E = \lambda_S \bar{t}$, from which ρ is absent because it enters only λ_D . Theorem 6.1 is the crossing point of the two components for the bystander: since $|\bar{t}| = |\tilde{t}_C|$, A depreciates against C if and only if $|\lambda_D| > |\lambda_S|$, i.e., $\rho/N > \rho_1^*$, where the impact disturbances load the two directions in the N -free ratio ρ/ρ_1^* and the factor N is the ratio of the adjustment eigenvalues of Lemma D.1, Nc for the zero-sum direction against c for the uniform one.

7.3 Real exchange rates

From (10) and (6),

$$d \log q_{Aj} = \nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}). \quad (46)$$

For a tariff vector t the price-index term depends only on the scale component, since $d \log P_{MA} = d\nu_A^E + \bar{t}$ and $d \log P_{Mj} = d\nu_A^E - \frac{N}{N-1} \nu_j$ for a partner that imposes no tariff, so

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - m \bar{t}, \quad dq_A^E = \left[\left(1 - \frac{mN}{N-1}\right) \lambda_S - m\right] \bar{t}, \quad d \log q_{Aj} - dq_A^E = \left(1 - \frac{mN}{N-1}\right) \lambda_D \tilde{t}_j. \quad (47)$$

The real effective rate is again scale-only, and real bilateral deviations are again discriminatory-only, attenuated by the pass-through factor. For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and both bilateral real exchange rates can be written as

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \quad (48)$$

The real exchange-rate response therefore combines the nominal response with a direct price-index effect that appreciates A in real terms. Setting $d \log q_{AC} = 0$ and using (32), for $m < 1/N$, gives

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_1^* + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)} > N \rho_1^*. \quad (49)$$

Averaging (48) gives

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \left(m < \frac{N-1}{N}\right). \quad (50)$$

Proposition 7.4 (Real rates). *Under (20), (i) the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N \rho_1^*$, and (ii) the REER appreciates for every ρ and N , $dq_A^E/d\tau < 0$.*

Both results follow from the direct price-index effect: the tariff raises A 's consumer prices at a given nominal exchange rate, making a bilateral real depreciation harder to obtain and reinforcing the effective appreciation.

8 Trade Wars and Preferential Tariffs

The decomposition extends to configurations in which several countries impose tariffs, since each tariffing country's vector decomposes as in Section 7 and the responses superpose. Under symmetry any country can serve as the reference country of Theorem 7.1, so for a tariffing country i with vector $t^{(i)}$ over its partners,

$$d \log e_{ij} = \lambda_S \bar{t}^{(i)} + \lambda_D \tilde{t}_j^{(i)}, \quad d \log e_{jk} = \lambda_D (t_k^{(i)} - t_j^{(i)}) \quad (j, k \neq i). \quad (51)$$

A tariffing country's scale component therefore moves its own bilateral rates one for one and cancels from every cross rate between its partners, because it moves those partners equally. Its discriminatory component enters every rate between differently treated countries.

8.1 Trade wars

Bilateral war. Adding the mirror shock $dt_{BA} = d\tau$ to the unilateral tariff gives

$$\left. \frac{d \log e_{AB}}{d\tau} \right|^w = 0, \quad \left. \frac{d \log e_{AC}}{d\tau} \right|^w = \frac{\rho - \rho_1^*}{D_N}, \quad \left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}. \quad (52)$$

Proposition 8.1 (Trade war threshold). *Under (20), in a symmetric trade war the belligerents' effective rates depreciate, and every bystander appreciates against both, if and only if $\rho > \rho_1^*$, independently of N .*

In the language of (51), B 's retaliation contributes to the A – C rate only through its discriminatory component, since its scale component appreciates B equally against A and C and cancels from e_{AC} :

$$\left. \frac{d \log e_{AC}}{d\tau} \right|^w = \underbrace{-\frac{\rho_1^*}{D_N}}_{A's \text{ scale}} + \underbrace{\frac{\rho}{ND_N}}_{A's \text{ discrimination}} + \underbrace{\frac{(N-1)\rho}{ND_N}}_{B's \text{ discrimination}} = \frac{\rho - \rho_1^*}{D_N}. \quad (53)$$

The two discriminatory contributions sum to ρ/D_N regardless of N , which is why the threshold falls from $N\rho_1^*$ to ρ_1^* . Retaliation adds a second discriminatory disturbance that singles out the same pair: it enters e_{AC} as a cross rate between B 's partners, with the full tariff difference $d\tau$ rather than the loading $d\tau/(N-1)$, so the discriminatory contribution is multiplied by N , which exactly offsets the factor N by which adjustment absorbs a single zero-sum disturbance (Lemma D.1), while no scale component is added to the A – C rate. On the A – B rate the two countries' components are mirror images and cancel exactly. At $\eta = 1$, $\rho_1^* = 1 - m < 1$.

Global retaliation. Suppose A raises its tariff by $d\tau$ on every partner and every partner retaliates with $d\tau$ on A .

Proposition 8.2 (Global retaliation). *Under (20), if A taxes all $N - 1$ partners by $d\tau$ and each partner taxes A by $d\tau$, then*

$$\frac{d \log e_{Aj}}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_1^*)}{D_N} \quad \forall j, \quad (54)$$

and all cross rates between partners are unchanged. A depreciates against every partner if and only if $\rho > \rho_1^$.*

A 's own tariff is pure scale and appreciates it by $(N-1)\rho_1^*/D_N$ against everyone. Each partner's retaliation is a unilateral tariff on A . Its scale component appreciates that partner against A by ρ_1^*/D_N and cancels from every other rate involving A , leaving a net scale appreciation of $(N-2)\rho_1^*/D_N$. Its discriminatory component singles out A and, summed over the $N-1$ retaliating partners, depreciates A by $(N-2)\rho/D_N$ against each. The threshold is again ρ_1^* , and at $N = 2$ the response vanishes by symmetry, as in the bilateral war.

8.2 Competing exporters

A standard model in the preferential-trade-agreement (PTA) literature has one importer and two competing exporters, following the tradition of Viner (1950) and Mundell (1964). Importer A holds only the numeraire good y , while exporters B and C hold good x . Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C . Let q_j denote the exporter price and p the domestic price. Market clearing requires

$$p = q_B + t_B = q_C + t_C, \quad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \quad M'_A < 0, \quad E'_j > 0, \quad (55)$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$, with u'_i equal to the local price. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1), \quad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (56)$$

Proposition 8.3 (Competing-exporter terms-of-trade effect). *A discriminatory tariff on B worsens B 's terms of trade and improves C 's terms of trade.*

8.3 Preferential tariffs as discrimination without scale

In our model, an improvement in C 's terms of trade is equivalent to a rise in e_{AC} by (12), so the two models can be compared directly: C 's terms of trade improve unconditionally in the competing-exporter model, while the same outcome requires $\rho > 3\rho_1^*$ in ours.

A preferential tariff is a discriminatory vector by construction, but in our model it also carries a scale component. Lowering A 's tariff on B alone is $t = -\tau \mathbf{e}_B$, the negative of the unilateral tariff, with $\bar{t} = -\tau/(N-1)$ and $\tilde{t}_C = \tau/(N-1)$. By Theorem 7.1 the effect on the non-member's currency is the compromise of Section 7.2 with the signs reversed: the scale component depreciates A against C because protection falls, the discriminatory component appreciates it because C is now taxed above the average, and C 's terms of trade worsen, the canonical direction, only if $\rho > N\rho_1^*$.

The difference between the two models is therefore the presence of the scale component. The competing-exporter model has two foreign suppliers of a common importable and quasilinear preferences, while our model has country-specific differentiated tradables and nested-CES preferences. Neither model is a special case of the other. The threshold at which discrimination overtakes scale at $N = 3$ can be written as

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B-C \text{ linkage}} + \underbrace{3\alpha_D(\eta-1)}_{\text{home switching}} - \underbrace{3\alpha_T(1-\alpha_D)}_{\text{openness offset}}. \quad (57)$$

The first term captures the effects operating through B - C trade: lower income in B reduces its purchases from C , while relative-price changes lead C to substitute toward cheaper goods from B . The second is A 's substitution toward its own tradable, and the final term works in the

opposite direction, since a larger import share gives the tariff more foreign expenditure to redirect, strengthening trade diversion and lowering the threshold.

In the language of Section 7, the canonical environment has no scale channel. Country A has no competing home tradable, so there is no home-switching margin, and quasilinearity with a traded numeraire removes the income and absorption responses. The overall level of protection is absorbed by the domestic price p rather than by an exchange-rate adjustment. What remains is the analogue of the discriminatory component alone, direct reallocation between the two foreign suppliers, which has an unambiguous sign because $M'_A < 0$ and $E'_j > 0$. A single component cannot produce a crossing, so the terms-of-trade effects of a preference are unconditional. In our model the same discriminatory force is present, but it must overcome the scale response that the level of protection induces, and the threshold measures how strong it must be.

This is an economic comparison rather than a formal nesting result. Quasilinear preferences cannot be obtained from the nested-CES demand system by a parameter restriction, and (57) is useful for interpreting the threshold rather than mapping one model into the other. The classic competing-exporter result isolates the trade-diversion force. Our model adds home substitution, income effects, and third-country trade, which can offset that force. The condition $\rho > 3\rho_1^*$ states how strong cross-origin substitution must be for trade diversion to dominate.

Read in these terms, the PTA literature studies the discriminatory component of trade policy, and the environments differ mainly in whether a scale channel exists to offset it. The broader PTA literature differs in which of these margins it retains. Competing-exporter models such as Bagwell and Staiger (1999) and Ornelas (2005) abstract from B – C trade and use quasilinear preferences, so the corresponding diversion effect is unopposed. Bloc and endowment models such as Krugman (1991) and Bond and Syropoulos (1996) instead abstract from home production of importables. Viner’s (1950) identical-goods case corresponds, in the language of our model, to the limit $\rho \rightarrow \infty$. Kemp and Wan (1976) consider a more general customs-union environment and do not provide the corresponding unconditional terms-of-trade sign result. Applied GTAP-class and quantitative trade models retain analogous general-equilibrium forces, but to our knowledge do not characterize the reversal threshold analytically.

9 Conclusion

The general equilibrium response to a tariff can be written as $(-J)^{-1}F$. The vector F gives the trade-balance disturbances created by the tariff at unchanged exchange rates, while J describes how exchange-rate changes restore trade balance.

At $N = 2$, this reduces to the textbook Marshall–Lerner result: an import tariff appreciates the tariffing country’s currency. With three or more countries, a discriminatory tariff also reallocates expenditure across foreign suppliers. The tariffing country still appreciates against the target, but its exchange rate against an untariffed bystander can move in either direction when cross-origin substitution is sufficiently strong.

Under symmetry, this bilateral ambiguity disappears after aggregation. The tariffing country’s NEER appreciates for every admissible ρ and N , even though individual bilateral rates may move

in opposite directions. The main distinction is therefore between substitution between home and foreign goods, which determines the sign of the effective exchange-rate response, and substitution across foreign suppliers, which determines how that adjustment is distributed across bilateral exchange rates.

Under symmetry these two margins are the two components of any tariff vector. The average tariff determines the effective rate through home switching and expenditure absorption, while deviations from the average determine bilateral rates around it through cross-origin substitution. Trade wars layer such components across countries, and the canonical preferential-tariff results are the discriminatory component operating without a scale channel to offset it.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (8), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ becomes $e m w_B L_B = m I_A / (1 + \tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}, \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_1^*}{D_2} \right|_{\eta=1}. \quad (58)$$

Thus, $e(\tau)$ is decreasing in τ , so the sign of the linearized response in (15) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D .

B Asymmetry

The disturbance-adjustment representation (18) and the multilateral Marshall–Lerner condition (20) do not require symmetry. The exact thresholds, the NEER cancellation, and the scale–discrimination decomposition in Sections 5–8 do. This appendix examines what survives without symmetry. Conditional on the multilateral Marshall–Lerner condition, the bilateral reversal result continues to hold at any balanced baseline where the tariffing country and the bystander each import from both foreign origins. The NEER result also survives to leading order within a structured asymmetric family.

A simple degree argument shows why the single-threshold results under symmetry do not extend directly to arbitrary trade networks. The elasticity ρ enters the share responses (25) only through $(1 - \rho)$, so every entry of J and F is linear in ρ . Each entry of $\text{adj}(-J)$ is a determinant of an $(N - 2) \times (N - 2)$ submatrix of $-J$ and is therefore a polynomial of degree at most $N - 2$ in ρ . It follows that each numerator $[\text{adj}(-J) F]_i$ in (18) is a polynomial of degree at most $N - 1$. Since $\det(-J) > 0$ under (20), a bilateral response can change sign only at a root of this numerator polynomial. Symmetry reduces the system to two unknowns and leaves numerators linear in ρ , producing the single thresholds in Proposition 5.1 and Theorem 6.1. At a general asymmetric baseline, the $N = 3$ numerator is already quadratic in ρ , as shown in (59), and for $N \geq 4$ a response may change sign more than once as ρ rises.

General asymmetric baseline. At any balanced baseline, (22)–(27) continue to hold with observed shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$, the bystander slope has denominator $\det(-J) > 0$ and a numerator that is quadratic in ρ (derived in the appendix of the companion IMF working paper by Lu and Milkov, forthcoming):

$$\text{sign } \frac{d \log e_{AC}}{d\tau} = \text{sign } (c_2 \rho^2 + c_1 \rho + c_0), \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C. \quad (59)$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. A positive numerator at large ρ determines the equilibrium sign only if the adjustment mechanism remains regular. The result is therefore conditional on the multilateral Marshall–Lerner condition (20) continuing to hold at the relevant values of ρ , which ensures $\det(-J) > 0$. Under this condition, $c_2 > 0$ whenever A and C each import from both foreign origins. Sufficiently strong cross-origin substitution then reverses e_{AC} at any such baseline. The threshold is the largest root and equals $3\rho_1^*$ at the symmetric point. Away from symmetry, there is no universal bilateral threshold.

Structured asymmetry. Let the target receive a multiple κ of its symmetric bilateral share. This scaling remains comparable across N because the symmetric share falls as $1/(N - 1)$. Assume the weight matrix is symmetric and balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N - 1}, \quad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N - 2}, \quad b_{CC'} = \frac{1 - 2b_{CA}}{N - 3}. \quad (60)$$

The reduced system again has two unknowns and can be solved in closed form. With NEER weights $w_{Aj} = b_{Aj}$, the response is a rational function of (ρ, N, κ) that reduces to $-\rho_1^*/D_N$ at $\kappa = 1$. Its numerator is $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$ whose leading coefficient is proportional to $(\kappa - 1)(N - 1 - \kappa)$.

Proposition B.1 (Asymmetric NEER). *Holding κ fixed as $N \rightarrow \infty$,*

$$N \frac{d\nu_A^E}{d\tau} \longrightarrow -\kappa \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\kappa \lim_{N \rightarrow \infty} \frac{N\rho_1^*}{D_N}. \quad (61)$$

To leading order, asymmetric exposure scales the symmetric NEER response by the relative importance of the target without changing its sign. At finite N , if $\kappa \geq 1$, all coefficients of Q are positive for large N , so the effective rate appreciates for every ρ . If $\kappa < 1$, Q has a single positive root,

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1 - \kappa} (1 + O(N^{-1})),$$

above which the NEER depreciates. When the target is under-weighted, depreciations against the many bystanders can outweigh the appreciation against B . As $\kappa \rightarrow 0$, $\rho_E^* \rightarrow N\rho_1^*$, so the reversal threshold remains of order N . In the language of Section 7, asymmetric weights make the zero-sum direction no longer an eigenvector of the response matrix, so the discriminatory component of the tariff leaks into the effective rate; the $\kappa < 1$ reversal is this leakage. The NEER result is therefore exact under symmetry and approximately robust within this structured asymmetric family. We do not claim robustness to arbitrary asymmetric trade networks.

C Correspondence across N

object	$N = 2$	$N = 3$	general N
ML condition (20)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 1)
sign rule	one impact, unconditional	weights $\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$\text{sign}(\sum_j \vartheta_{ij} F_j), \vartheta \geq 0$
bilateral threshold	none (no by-stander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree- $(N - 1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$	$\rho_q^*(N)$ (49)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$, sign invariant under symmetry
scale, discrimination eigenvalues (40)	$-\rho_1^*/D_2, -$	$-2\rho_1^*/D_3, -2\rho/(3D_3)$	$-(N - 1)\rho_1^*/D_N, -(N - 1)\rho/(ND_N)$
war threshold	$-$	ρ_1^*	ρ_1^* , independent of N

D Proofs

Throughout, $m \equiv \alpha_T(1 - \alpha_D)$, $P \equiv \rho + \alpha_D(\eta - 1)$, and all derivatives are evaluated at the balanced free-trade baseline of Section 5.1.

Proof of Proposition 3.1. The collected condition (14) and the slope (15) are derived in Section 3. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1 - \eta)\alpha_D = 1 + \alpha_D(\eta - 1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta - 1)$. Differentiating $\text{TB}_A(e, \tau) = X(1/e) - eM((1 + \tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives $\partial \text{TB}_A / \partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta - 1) + 1 = D_2$, so $D_2 > 0$ is exactly the condition $\varepsilon_X + \varepsilon_M > 1$. Since $m\rho_1^* > 0$ by (13), the slope $-\rho_1^*/D_2$ is negative. $\square$

Proof of Proposition 4.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (22) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i , by the same factor. Thus, TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \tilde{J}_{il} = \text{TB}_i$, which is zero at the balanced baseline, so $\sum_l \tilde{J}_{il} = 0$ for all i . Equivalently, in (27), a common shift in all ν_l affects only the income terms, whose export

and import contributions cancel by baseline balance. Summing (9) over i gives $\sum_i \text{TB}_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$ for all l . Under Assumption 1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives

$$|J_{ii}| = \sum_{l \neq i, A} J_{il} + \tilde{J}_{iA} \geq \sum_{l \neq i, A} J_{il}, \quad (62)$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$.

Thus, $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \text{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \geq 0.$$

Nonsingularity follows from $r < 1$. Moreover, $\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0$: $\det D > 0$, and the eigenvalues $1 - \lambda$ of $I - D^{-1}B$ satisfy $|\lambda| \leq r < 1$, so real eigenvalues are positive and complex eigenvalues occur in conjugate pairs with positive products. Finally, the Gershgorin discs of $-J$ are centered at $|J_{ii}| > 0$ with radii no larger than $|J_{ii}|$. They therefore lie in the closed right half-plane and can intersect the imaginary axis only at the origin, which nonsingularity excludes. Hence $\text{Re } \lambda(-J) > 0$, or equivalently $\text{Re } \lambda(J) < 0$. $\square$

Proof of Corollary 4.3. The result follows directly from (18): $d\nu/d\tau = (-J)^{-1}F$, with $(-J)^{-1} = (\vartheta_{ij}) \geq 0$ elementwise by Proposition 4.2. $\square$

Lemma D.1 (Full Jacobian at the symmetric baseline). *At the symmetric balanced baseline of Section 5.1, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad \tilde{J}_{ii} = -\frac{m}{N-1} D_N, \quad D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1. \quad (63)$$

Proof. Set $dt_{jk} = 0$, so $dp_{ij} = \nu_j - \nu_i$ and $d \log \tilde{I}_i = \nu_i$. Writing $S \equiv \sum_j \nu_j$, (23) gives

$$d \log P_{M_k} = \frac{S - N\nu_k}{N-1}, \quad \frac{\partial d \log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side, $\sum_{j \neq i} (dp_{ij} - d \log P_{M_i}) = 0$, so the $(1-\rho)$ terms cancel when (25) is summed over $j \neq i$. Thus, $\sum_{j \neq i} d \log s_{ij} = (N-1)(1-\eta)\alpha_D d \log P_{M_i}$. Substituting into (27) gives

$$d \text{TB}_i = \frac{m}{N-1} \left[\sum_{k \neq i} (d \log s_{ki} + \nu_k) - (N-1)(1-\eta)\alpha_D d \log P_{M_i} - (N-1)\nu_i \right].$$

Fix $l \neq i$ and differentiate term by term, using $\sum_{k \neq i} \delta_{kl} = 1$ and $\sum_{k \neq i} (1 - N\delta_{kl}) = -1$:

$$\frac{\partial}{\partial \nu_l} \sum_{k \neq i} d \log s_{ki} = (1 - \rho) \left(-1 + \frac{1}{N-1} \right) - \frac{(1 - \eta)\alpha_D}{N-1} = -(1 - \rho) \frac{N-2}{N-1} - \frac{(1 - \eta)\alpha_D}{N-1}.$$

The export income terms contribute $\partial_{\nu_l} \sum_{k \neq i} \nu_k = 1$, the import-side price-index term contributes $-(1 - \eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} \left[(N-1) - (1 - \rho)(N-2) + N\alpha_D(\eta-1) \right] = \frac{m}{(N-1)^2} D_N.$$

The diagonal follows from (19): $\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -\frac{m}{N-1}D_N$. As checks, at $N = 2$ this gives $\tilde{J}_{AB} = mD_2$, the coefficient in (14); deleting row and column A at $N = 3$ reproduces J in (33); and collapsing the $N - 2$ identical bystanders reproduces (31). $\square$

Proof of Proposition 5.1. At $N = 3$, (22)–(23) give

$$d \log P_{MA} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{MB} = \frac{1}{2}\nu_C - \nu_B, \quad d \log P_{MC} = \frac{1}{2}\nu_B - \nu_C. \quad (64)$$

Also, $d \log \tilde{I}_A = \frac{m}{2}d\tau$, $d \log \tilde{I}_B = \nu_B$, and $d \log \tilde{I}_C = \nu_C$. The deviations of bilateral prices from the corresponding import indices are

$$\begin{aligned} dp_{AB} - d \log P_{MA} &= \frac{1}{2}(\nu_B - \nu_C + d\tau), & dp_{AC} - d \log P_{MA} &= \frac{1}{2}(\nu_C - \nu_B - d\tau), \\ dp_{BA} - d \log P_{MB} &= -\frac{1}{2}\nu_C, & dp_{BC} - d \log P_{MB} &= \frac{1}{2}\nu_C, \\ dp_{CA} - d \log P_{MC} &= -\frac{1}{2}\nu_B, & dp_{CB} - d \log P_{MC} &= \frac{1}{2}\nu_B. \end{aligned}$$

The share changes then follow from (25). With flow values $m/2$, (27) gives

$$\begin{aligned} \frac{2}{m} d \text{TB}_B &= (d \log s_{AB} + \frac{m}{2}d\tau - d\tau) + (d \log s_{CB} + \nu_C) - (d \log s_{BA} + \nu_B) - (d \log s_{BC} + \nu_B), \\ \frac{2}{m} d \text{TB}_C &= (d \log s_{AC} + \frac{m}{2}d\tau) + (d \log s_{BC} + \nu_B) - (d \log s_{CA} + \nu_C) - (d \log s_{CB} + \nu_C). \end{aligned}$$

For example, the coefficient on ν_B in $\frac{2}{m}d \text{TB}_B$ is

$$\underbrace{\frac{1}{2}[(1 - \rho) + (1 - \eta)\alpha_D]}_{d \log s_{AB}} + \underbrace{\frac{1}{2}[(1 - \rho) + (1 - \eta)\alpha_D]}_{d \log s_{CB}} + \underbrace{2(1 - \eta)\alpha_D - 2}_{-d \log s_{BA} - d \log s_{BC} - 2\nu_B} = -(3\alpha_D(\eta-1) + \rho + 1) = -D_3.$$

Collecting the remaining coefficients gives (33); the replication package verifies the full collection symbolically. Inverting as in (34) gives (35). With $D_3 > 0$, the numerator for e_{AB} , $-(\rho + 3\rho_1^*)$, is negative and the numerator for e_{BC} , 2ρ , is positive for all admissible parameters. The bystander response satisfies $\text{sign}(d \log e_{AC}/d\tau) = \text{sign}(\rho - 3\rho_1^*)$. $\square$

Proof of Corollary 5.2. Setting $\rho = \eta = \sigma$ in $3\rho_1^* = 3[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)]$ gives

$$\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T).$$

If $\alpha_D \geq 1/3$, both terms are nonnegative, with equality in both only at $\alpha_D = 1/3$ and $\alpha_T = 1$. Reversal requires $\rho > \rho_3^*$, or equivalently $\sigma > \rho_3^*$, which is impossible whenever $\rho_3^* - \sigma \geq 0$. At $\alpha_D = 1/3$ and $\alpha_T = 3/4$, $\rho_3^* - \sigma = \frac{1}{2}$ and $D_3 = 3 \cdot \frac{1}{3}(\sigma - 1) + \sigma + 1 = 2\sigma$, so $d \log e_{AC}/d\tau = (\sigma - \rho_3^*)/(3D_3) = -1/(12\sigma)$. $\square$

Proof of Theorem 6.1. The equilibrium is unique by Proposition 4.2. Since the system is invariant to permutations of the $N - 2$ bystanders, the unique solution is also permutation-invariant, so all bystanders have the same response ν_C . The system therefore reduces to the conditions for B and one representative bystander. From (30),

$$\begin{aligned} dp_{AB} - d \log P_{M_A} &= \frac{N-2}{N-1}(\nu_B + d\tau - \nu_C), & dp_{AC} - d \log P_{M_A} &= \frac{1}{N-1}(\nu_C - \nu_B - d\tau), \\ dp_{BA} - d \log P_{M_B} &= -\frac{N-2}{N-1}\nu_C, & dp_{BC} - d \log P_{M_B} &= \frac{1}{N-1}\nu_C, \\ dp_{CA} - d \log P_{M_C} &= -\frac{1}{N-1}[(N-3)\nu_C + \nu_B], & dp_{CB} - d \log P_{M_C} &= \frac{1}{N-1}[(N-2)\nu_B - (N-3)\nu_C], \\ dp_{CC'} - d \log P_{M_C} &= \frac{1}{N-1}(2\nu_C - \nu_B). \end{aligned}$$

Using (25) and flow values $m/(N-1)$, the two trade-balance conditions are

$$\begin{aligned} \frac{N-1}{m} d \text{TB}_B &= (d \log s_{AB} + \frac{m}{N-1}d\tau - d\tau) + (N-2)(d \log s_{CB} + \nu_C) - (d \log s_{BA} + \nu_B) - (N-2)(d \log s_{BC} + \nu_B) \\ \frac{N-1}{m} d \text{TB}_C &= (d \log s_{AC} + \frac{m}{N-1}d\tau) + (d \log s_{BC} + \nu_B) + (N-3)(d \log s_{C'C} + \nu_C) \\ &\quad - (d \log s_{CA} + \nu_C) - (d \log s_{CB} + \nu_C) - (N-3)(d \log s_{CC'} + \nu_C). \end{aligned}$$

Collecting coefficients as in the proof of Proposition 5.1, and multiplying both rows by a further factor $N-1$ so that every coefficient is a multiple of D_N , gives (31); the replication package verifies this collection symbolically. The counting matrix has determinant $2(N-1) - (N-2) = N$, so inversion gives (32), which reduces to (35) at $N=3$ and (15) at $N=2$. Under (20), $D_N > 0$. The numerator for e_{AB} , $-[N\rho_1^* + (N-2)\rho]$, is therefore negative, while that for e_{BC} is positive. Finally, $\text{sign}(d \log e_{AC}/d\tau) = \text{sign}(\rho - N\rho_1^*)$. $\square$

Proof of Theorem 6.2. With symmetric weights $w_{Aj} = 1/(N-1)$, (32) gives

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \frac{-[N\rho_1^* + (N-2)\rho] + (N-2)[\rho - N\rho_1^*]}{(N-1)ND_N} = -\frac{\rho_1^*}{D_N}.$$

The terms in ρ cancel. By (13), $\rho_1^* = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T) > 0$ for $\alpha_D, \eta > 0$ and $\alpha_T \leq 1$. Since $D_N > 0$ under (20), the response is negative for every $N \geq 2$ and every admissible ρ . $\square$

Proof of Theorem 7.1. By linearity, $\nu = \sum_k t_k r^{(k)}$, where $r^{(k)}$ is the response to a unit tariff on partner k alone. By the symmetry of the partners, $r^{(k)}$ is the unilateral response (32) with B relabeled as k : its k -th entry is α and every other entry is β , which is (39), i.e., $\nu = R\mathbf{t}$ with $R = (\alpha - \beta)I + \beta\mathbf{1}\mathbf{1}'$. Since $\mathbf{1}'\mathbf{1} = N-1$, $R\mathbf{1} = [\alpha + (N-2)\beta]\mathbf{1} = \lambda_S\mathbf{1}$, and for any v with $\mathbf{1}'v = 0$, $Rv = (\alpha - \beta)v = \lambda_D v$. Substituting α and β gives (40). Writing $t = \bar{t}\mathbf{1} + \tilde{t}$ with $\mathbf{1}'\tilde{t} = 0$ gives $\nu = \lambda_S\bar{t}\mathbf{1} + \lambda_D\tilde{t}$. Averaging with weights $1/(N-1)$ gives $d\nu_A^E = \lambda_S\bar{t}$ because $\mathbf{1}'\tilde{t} = 0$, and subtracting gives the deviations. Finally, $d \log e_{jk} = \nu_k - \nu_j = \lambda_D(\tilde{t}_k - \tilde{t}_j) = \lambda_D(t_k - t_j)$. Under (20), $D_N > 0$ by Section 5.1, so $\lambda_S, \lambda_D < 0$. $\square$

Proof of Corollary 7.2. (i) From (41), $\nu_j - \nu_k = \lambda_D(t_j - t_k)$ with $\lambda_D < 0$. (ii) $\nu_j > 0$ iff $\lambda_S \bar{t} + \lambda_D(t_j - \bar{t}) > 0$. Substituting (40) and dividing by $(N-1)/D_N > 0$ gives $-\rho_1^* \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0$, i.e., $\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t}$; dividing by $\bar{t} > 0$ gives the second form. The right-hand side $1 - N\rho_1^*/\rho$ is positive iff $\rho > N\rho_1^*$, and $t_j/\bar{t} \geq 0$, so the condition is infeasible otherwise. $\square$

Proof of Corollary 7.3. With $t_j = \tau$ on the k targets and zero otherwise, $\bar{t} = k\tau/(N-1)$, $\tilde{t}_B = \tau(N-1-k)/(N-1)$ for a target, and $\tilde{t}_C = -k\tau/(N-1)$ for a bystander. Substituting into (41), $\nu_C = \frac{k\tau}{N-1}(\lambda_S - \lambda_D) = \frac{k\tau}{ND_N}(\rho - N\rho_1^*)$, $\nu_B = \lambda_S \frac{k\tau}{N-1} + \lambda_D \frac{(N-1-k)\tau}{N-1}$, and $d\nu_A^E = \lambda_S k\tau/(N-1)$, which are (43). The sign of ν_C is that of $\rho - N\rho_1^*$ for every k . At $k = N-1$ there is no bystander and $\tilde{t} = 0$, giving (44). $\square$

Proof of Proposition 8.1. Let primes denote responses to the mirror shock $dt_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (32) gives $d \log e'_{BA} = -[N\rho_1^* + (N-2)\rho]/(ND_N)$ and $d \log e'_{BC} = (\rho - N\rho_1^*)/(ND_N)$. Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d \log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{AC} = d \log e'_{AB} + d \log e'_{BC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity, the war response is the sum of the original and mirror-shock responses:

$$d \log e_{AB}|^w = 0, \quad d \log e_{AC}|^w = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

By symmetry between A and B , $d \log e_{BC}|^w = d \log e_{AC}|^w$. The war NEER is therefore $\frac{1}{N-1}[0 + (N-2)(\rho - \rho_1^*)/D_N]$. With $D_N > 0$, all three statements have the sign of $\rho - \rho_1^*$, independently of N . In the components of (51), B 's vector has $\tilde{t}^{(B)} = d\tau/(N-1)$ and $\tilde{t}_C^{(B)} = -d\tau/(N-1)$, and its contribution to e_{AC} is $\lambda_D(t_C^{(B)} - t_A^{(B)}) = -\lambda_D d\tau = (N-1)\rho d\tau/(ND_N)$, which is (53). $\square$

Proof of Proposition 8.2. Superpose the N tariff vectors using (51). A 's vector is uniform, $\tilde{t}^{(A)} = d\tau$ and $\tilde{t}_A^{(A)} = 0$, contributing $\lambda_S d\tau$ to every $d \log e_{Aj}$. Partner j 's vector taxes only A : $\tilde{t}^{(j)} = d\tau/(N-1)$, $\tilde{t}_A^{(j)} = \frac{N-2}{N-1}d\tau$, and $\tilde{t}_k^{(j)} = -d\tau/(N-1)$ for $k \neq A, j$. It contributes $-\left[\lambda_S + (N-2)\lambda_D\right]\frac{d\tau}{N-1}$ to $d \log e_{Aj} = -d \log e_{jA}$, and $\lambda_D(t_k^{(j)} - t_A^{(j)}) = -\lambda_D d\tau$ to each cross rate $d \log e_{Ak}$, $k \neq j$. Summing A 's own vector, partner j 's vector, and the $N-2$ other partners' vectors,

$$d \log e_{Aj} = \lambda_S d\tau - \left[\lambda_S + (N-2)\lambda_D\right]\frac{d\tau}{N-1} - (N-2)\lambda_D d\tau = \frac{N-2}{N-1}(\lambda_S - N\lambda_D)d\tau = \frac{(N-2)(\rho - \rho_1^*)}{D_N}d\tau.$$

The expression is the same for every j , so it equals the NEER response, and all cross rates between partners are zero. The same result follows by direct superposition of (44) with the relabeled unilateral responses in (32). $\square$

Proof of Proposition 7.4. From (10) and (6), only tradable prices move, so $d \log q_{Aj} = \nu_j + \alpha_T(d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}})$, with $d \log P_{T_i}^{\text{own}} = (1 - \alpha_D)d \log P_{M_i}$. The indices in (30) are already expressed in own currency. Hence

$$d \log q_{AB} = \nu_B + m \left[\frac{(N-2)\nu_C}{N-1} - \nu_B - \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1} \right] = \left(1 - \frac{mN}{N-1} \right) \nu_B - \frac{m}{N-1} d\tau.$$

The same calculation for $j = C$ gives (48), since the corresponding bracket reduces to $-\frac{N\nu_C + d\tau}{N-1}$. For a general tariff vector, $d \log P_{MA} = \frac{1}{N-1} \sum_j (\nu_j + t_j) = d\nu_A^E + \bar{t}$ and, since partners impose no tariffs, $d \log P_{M_j} = \frac{1}{N-1} [\sum_{k \neq A} \nu_k - N\nu_j] = d\nu_A^E - \frac{N}{N-1} \nu_j$, so the bracket is $-\frac{N}{N-1} \nu_j - \bar{t}$, which gives (47); the unilateral case $\bar{t} = d\tau/(N-1)$ is (48). Setting $d \log q_{AC} = 0$ and substituting ν_C from (32) gives

$$[(N-1) - mN] \frac{\rho - N\rho_1^*}{N} = mD_N(\rho).$$

Because $D_N(\rho) = N\alpha_D(\eta-1) + (N-2)\rho + 1$ is linear in ρ , solving gives (49), using $(N-1) - mN - mN(N-2) = (N-1)(1-mN)$. If $m < 1/N$, the factor $(N-1) - mN$ is positive, and $D_N(\rho_q^*) > 0$ for $\eta \geq 1$. Hence $\rho_q^* > N\rho_1^*$, proving part (i). Averaging (48) over j and substituting (37) gives (50). Under (20), both terms are negative for $m < (N-1)/N$, proving part (ii). $\square$

Proof of Proposition 8.3. The result follows from (56). Total differentiation of market clearing gives $dp/dt_B \in (0, 1)$, and $M'_A < 0$ and $E'_j > 0$ then imply the stated signs of dq_B/dt_B and dq_C/dt_B . $\square$

Proof of Proposition B.1. Under the weights in (60), the bystanders remain interchangeable. By the same uniqueness and symmetry argument used in Theorem 6.1, the system again reduces to two unknowns (ν_B, ν_C) . The corresponding flow values are $m\beta$, $m\gamma$, and $m\xi$, where $\beta = \kappa/(N-1)$, $\gamma = (1-\beta)/(N-2)$, and $\xi = (1-2\gamma)/(N-3)$. Write $\nu_B = b/N$ and $\nu_C = c/N$, holding $d\tau$ fixed, and expand the two equilibrium conditions in $1/N$ at fixed (κ, ρ) . To leading order, $\beta \rightarrow \kappa/N$, $(N-2)\gamma \rightarrow 1$, and

$$d \log P_{MA} = \frac{\kappa d\tau + c}{N} + O(N^{-2}), \quad d \log P_{MB} = \frac{c-b}{N} + O(N^{-2}), \quad d \log P_{MC} = O(N^{-2}).$$

Substituting into (25) and the flow sums, and multiplying the B and C conditions by N/m and $N(N-2)/m$, respectively, gives the limiting system

$$P(b-c) = -\kappa\rho d\tau, \quad P(2c-b) = \kappa(\rho - \rho_1^*) d\tau, \quad P \equiv \rho + \alpha_D(\eta-1) > 0 \quad \text{for } \eta \geq 1.$$

The first equation comes from the tariff term $\beta[(1-\rho)-1]d\tau = -\beta\rho d\tau$ and the relative share responses between B and the bystanders. The second comes from A 's diversion toward C net of C 's exposure to B , whose tariff coefficient is $\kappa[\rho-1+(1-\eta)\alpha_D+m] = \kappa(\rho-\rho_1^*)$. The limiting matrix $P \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ has determinant $P^2 > 0$, so the solution of the exact system converges to the solution of the limiting system. Adding the two equations gives $c = -\kappa\rho_1^* d\tau/P$. Since the NEER weights satisfy $w_{AB} = \beta = O(N^{-1})$ and $w_{AC} = \gamma$, with $(N-2)\gamma \rightarrow 1$,

$$N \frac{d\nu_A^E}{d\tau} = N[\beta\nu_B + (N-2)\gamma\nu_C]/d\tau \rightarrow \frac{c}{d\tau} = -\frac{\kappa\rho_1^*}{\rho + \alpha_D(\eta-1)}.$$

This equals $-\kappa \lim_N N\rho_1^*/D_N$ because $D_N/N \rightarrow P$. $\square$